MIL-C-13540(Ord)
13 July 1954

SUPERSEDING
PA-PD-89 (Rev 1)
29 April 1954

MILITARY SPECIFICATION

CLOTH, CARTRIDGE, HIGH TENACITY DIRECT SPUN VISCOSE RAYON

1. SCOPE

1.1 This specification covers one type of high tenacity direct spun viscose rayon cloth used in manufacturing cartridge bags for artillery propelling charges.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, and all specifications, and drawings referenced thereon, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

CCC-T-191	-Textiles; Test Methods
DDD-S-751	-Stitches, Seams And Stitching

MILITARY

MIL-G-2550	-General Specification For Ammunition Except Small Arms Ammunition
MIL-P-10025	-Packing And Marking For Domestic Shipment of Inert Ammunition Components; General Specification For

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Rayon cartridge cloth shall be woven from direct spun high tenacity, viscose rayon yarn. The warp yarns only, shall be sized with gelatin applied from an aqueous bath not to exceed 165°F, containing

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3 percent (by weight of bath) of an undegraded gelatin and 0.25 percent (by weight of bath) of para-nitrophenol.

3.2 Appearance.-The appearance of the cloth shall be uniform and free from visible imperfections.

3.3 Width.-The width of the cloth shall be as specified in the contract or purchase order.

3.4 Weave.-Unless otherwise specified, the weave shall be a plain single weave.

3.5 Weight.-The weight of the cloth shall be 3.2 ounces per square yard maximum, and 2.8 ounces per square yard minimum when tested as specified in 4.5.

3.6 Breaking strength

3.6.1 Warp.-The breaking strength of the warp yarns shall be a minimum of 50 pounds when tested as specified in 4.6.

3.6.2 Fill.-The breaking strength of the fill yarns shall be a minimum of 50 pounds when tested as specified in 4.6.

3.7 Slippage

3.7.1 Warp.-The slippage value of the warp yarns shall be a minimum of 135 when tested as specified in 4.7.

3.7.2 Fill.-The slippage value of the fill yarns shall be a minimum of 125 when tested as specified in 4.7.

3.8 Stretch

3.8.1 Warp.-The stretch of the warp yarns shall be a maximum of 8.5 percent when tested as specified in 4.8.

3.8.2 Fill.-The stretch of the fill yarns shall be a maximum of 8.5 percent when tested as specified in 4.8.

3.9 Ash.-The ash shall be 1.0 percent maximum when tested as specified in 4.9.

3.10 pH of water extract.-The pH of water extract shall be 5.5 minimum to 7.5 maximum when tested as specified in 4.10.

3.11 Acidity or alkalinity

3.11.1 Alkalinity.-There shall be no alkalinity determined as sodium carbonate, when tested as specified in 4.11.

3.11.2 Acidity.-The acidity determined as acetic acid shall be not more than 0.1 percent when tested as specified in 4.11.

3.12 Thread Count

3.12.1 Warp.-The thread count of the warp yarns shall be from 60 minimum to 65 maximum when tested as specified in 4.12.

3.12.2 Fill.-The thread count of the fill yarns shall be from 60 minimum to 65 maximum when tested as specified in 4.12.

3.13 Ether soluble material.-There shall be a maximum of 0.5 percent ether soluble material when tested as specified in 4.13.

3.14 Objectionable sizing.-The cloth shall pass the objectionable sizing test specified in 4.14.

3.15 Workmanship.-The finished cloth shall be uniformly and closely woven and shall be free from weighting materials.

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot.-A lot shall consist of cloth manufactured by one manufacturer, subjected to the same manufacturing conditions and process and submitted for inspection at the same time.

4.2 Sampling.-Five representative primary samples shall be selected from the bales comprising the lot and all tests shall be performed on each sample. Failure of any sample to comply with all requirements shall be cause for rejection of the lot.

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4.3 Inspection.-Inspection shall be as specified in Specification MIL-G-2550.

4.4 Width.-The width shall be determined as directed by Specification CCC-T-191, Method 5020, to determine compliance with 3.3.

4.5 Weight.-The weight shall be determined as directed by Specification CCC-T-191, Method 5040, to determine compliance with 3.5.

4.6 Breaking strength.-The breaking strength shall be determined concurrently with the slippage value (see 4.7) as directed by Specification CCC-T-191, Method 5100, to determine compliance with 3.6.1 and 3.6.2.

4.7 Slippage value

4.7.1 Apparatus.-For the determination of the slippage value, a pendulum type testing machine shall be used as specified in Specification CCC-T-191, Method 5100, to determine compliance with 3.7.1 and 3.7.2.

4.7.2 Preparation of specimens.-Specimens, 4 inches wide and 16.0 to 16.5 inches long, shall be taken for test. Two sets of five specimens each are required, one set for warp slippage value, having the short dimension parallel to the warp yarns, and the other set for the fill slippage value having the short dimension parallel to the filling yarns. No two specimens for warp slippage shall contain the same warp yarns and no two specimens for the fill slippage shall contain the same filling yarns. Unless otherwise specified, no specimen shall be taken nearer the selvage than one-tenth the width of the material. The specimens shall be prepared by cutting strips of cloth of the number and in the directions specified above, approximately 4.5 inches wide and 17 inches long. Any cut or broken threads shall then be raveled from the strip of cloth until only threads remain in the length-wise direction of the cloth which are approximately the same length as the cloth strip and only threads remain in the short direction which are approximately as long as the width of the cloth strip.

Sufficient yarns shall then be raveled from both the long and short dimensions of the strip of cloth until the specimen is 4 inches wide by 16 to 16½ inches long. Any other convenient method of preparing the specimen may be used provided it doesn't affect the physical properties of the cloth.

A test specimen shall be folded back on itself so that the distance from the fold to one edge, parallel to the short dimension, measures approximately 5 inches and to the opposite edge at least 10 inches. The fold must be parallel to the crosswise yarns in the specimen. At 0.38 inch from the fold, a seam shall be sewn parallel to the crosswide yarns of the specimen. The seam shall be of the type designated as 301-SSa-1 by Specification DDD-S-751. (The seam SSa-1 shall be made with 1 row of stitch, 12 stitches per inch, Type 301.) The seam shall be made with a suitable needle and a thread of sufficient strength so that it will not break when the seam is tested for elongation as directed below (see 6.2). The specimens shall be conditioned for at least 24 hours at 21° ± 2°C and 65 ± 2 percent relative humidity.

4.7.3 Procedure.-The distance between the jaws of the test machine shall be adjusted to 3 inches. A 6-ounce clamp shall be attached to the 10-inch length of the specimen, parallel to and at a distance of 3 inches from the seam described in 4.7.2. The clamp can be assembled from a metallic photographic film clip approximately 5 inches long, a brass or steel rod 6 inches long and 3/16 inch in diameter or brass or steel weights, equal in weight, attached to the 6-inch rod to bring the total weight of the assembly to 6 ounces. The test specimen shall be placed with the

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10-inch portion uppermost, symetrically in the upper jaws of the test machine so that the long direction of the specimen is parallel to the direction of application of the load. The specimen shall be clamped in the upper jaws so that the 6-ounce clamp described above is approximately 1/4 inch below the lower jaws of the test machine. The test machine shall be reset so that a zero load is shown on the graphic record. The 6-ounce clamp shall be checked to be certain that it is supported only by the test specimen and that a uniform tension of 6 ounces is applied to the specimen. The lower jaws of the test machine shall be fastened to the specimen, care being taken that the same yarns gripped in the lower jaws are gripped in the upper jaws. The apparatus shall be arranged to obtain a tension-stretch curve up to the breaking strength of the specimen, the stretch being indicated in inches by the vertical component of the curve and the tension being indicated in pounds by the horizontal component. The machine shall be started and allowed to apply tension until the specimen breaks. This graphic record shall be designated as the fabric-stretch curve. The same specimen shall be inserted in the jaws of the machine so that the seam is in a position parallel to the jaws and midway between the upper and lower jaws. The specimen shall be clamped in the jaws, with tension being applied by means of the 6-ounce clamp, as directed above, before tightening the lower jaws. The apparatus shall be arranged so that a graphic record of this test will be made on the same chart and will start at the same point as the fabric-stretch curve. The machine shall be started and allowed to apply tension to the specimen until either the cloth breaks or the seam fails. This graphic record shall be designated as the seam-stretch curve. The tension shall be noted at which the vertical distance between the seam-stretch curve and the fabric-stretch curve is equal to the sum of:

a. the vertical distance between these curves when the tension is 1 pound, and
b. a vertical distance on the chart equivalent to 0.25 inch on the cloth,

to determine compliance with 3.7.1 and 3.7.2. The slippage shall be calculated as follows:

$$\text{Slippage value} = \frac{100A}{25}$$

where:

A = Tension at which the difference between the seam-stretch curve and the fabric-stretch. curve is 0.25 inch plus the difference between these curves when the tension is 1 pound.

The slippage value in the warp direction shall be considered as the average of five results obtained for the warp slippage specimens and the slippage value in the fill direction as the average of five results obtained for the fill slippage specimens.

4.8 Stretch.-The percent of stretch shall be determined concurrently with the slippage value and shall be calculated as shown below from measurements of the fabric-stretch curve of the graphic record. The percent of stretch of the warp yarns is obtained from the fabric-stretch curve obtained when the slippage value of the fill yarns is determined. The percent of stretch of the fill yarns is obtained from fabric-stretch

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curve obtained when the slippage value of the warp yarn is determined. The percent of stretch in each direction shall be considered as the average of five values determined in that direction and shall be calculated to determine compliance with 3.8.1, and 3.8.2 as follows:

$$\text{Percent of stretch} = \frac{100\ (A-B)}{3.00 + B}$$

where:

A = Elongation in inches of the portion of the specimen tested when the tension is 20 pounds as read on the fabric-stretch curve.
B = Elongation in inches of the portion of the specimen tested when the tension is 1 pound as read on the fabric-stretch curve.

The elongation of the specimen is represented by the vertical component of the fabric-stretch curve at the required tension.

4.9 Ash.-Parts from each of the pieces comprising the sample shall be cut, so as to obtain a total of approximately 2 gm, transferred to a tared porcelain crucible, and moistened with concentrated nitric acid. The crucible shall be heated on a steam bath for 1 hour and then heated carefully over a flame so as to avoid any mechanical loss. When the contents of the crucible have been charred, it shall be heated to dull redness until all carbonaceous matter has been burned off. The crucible shall be cooled in a desiccator, and weighed. The percent of ash in the sample shall be calculated to determine compliance with 3.9 as follows:

$$\text{Percent ash} = \frac{100\ (A-B)}{W}$$

where:

A = Weight of crucible and ignited residue.
B = Weight of crucible empty.
W = Weight of sample.

4.10 pH of water extract.-The pH value of the cloth shall be determined as directed by Specification CCC-T-191, Method 2811 to determine compliance with 3.10.

4.11 Acidity or alkalinity.-The water extract obtained in the determination of pH shall be decanted into a 100-ml beaker forcing as much absorbed water as possible out of the cloth by pressing the cloth against the side of the beaker with the aid of a glass stirring rod. Should the pH of the water extract be 7.0 or greater, the solution shall be titrated with approximately 0.1N hydrochloric acid of known normality, using methyl red as the indicator to a faint reddish tint. Any alkalinity shall be calculated to percent sodium carbonate to determine compliance with 3.11.1 as follows:

$$\text{Percent sodium carbonate} = \frac{5.3\ VN}{W}$$

where:

V = ml of hydrochloric acid solution required.

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N = Normality of the hydrochloric acid solution.
W = Weight of sample.

If the pH of the water extract is 6.9 or less, the solution shall be titrated with approximately 0.1N sodium hydroxide solution of known normality using phenolphthalein indicator to the first permanent pink color. Any acidity shall be calculated to percent acetic acid to determine compliance with 3.11.2 as follows:

$$\text{Percent of acetic acid} = \frac{6.0\ VN}{W}$$

where:

V = ml of sodium hydroxide solution required.
N = normality of sodium hydroxide solution.
W = Weight of sample.

4.12 Thread count per inch.-The thread count shall be determined as directed by Specification CCC-T-191, Method 5050 to determine compliance with 3.12.1 and 3.12.2.

4.13 Ether soluble material.-Ten pieces of cloth, weighing approximately 0.5 gm each shall be cut from various portions of the sample. These pieces shall be weighed together, and transferred to the extraction chamber of a Soxhlet extraction apparatus having ground-glass connections. The extraction chamber shall be assembled to a tared receiving flask and ether transferred to the extraction chamber until the sample is covered and the ether siphons over into the flask. Approximately 25 ml more ether shall be added to the extraction chamber, a condenser assembled to it and the assembled apparatus placed on a steam bath. The extraction shall be allowed to proceed for 1.5 hours. Heating shall be discontinued at a time when the ether content of the extraction chamber is nearly sufficient to cause the ether to siphon over into the receiving flask. The flask containing the oily ether extract shall be heated on a steam bath until the ether is removed. The flask and residue shall be dryed in an oven at 100° ± 2°C for 1 hour, cooled in a desiccator, and weighed. The weight of residue shall be calculated to percent ether extract in the sample to determine compliance with 3.13 as follows:

$$\text{Percent ether soluble material} = \frac{100A}{W}$$

where:

A = Weight of residue.
W = Weight of sample.

4.14 Objectionable sizing

4.14.1 Apparatus.-A Mackey test apparatus (see Figure 1) shall be prepared consisting of a cylindrical water jacketed metal oven of the following dimensions: Outside, 8 inches high and 6 inches in diameter; inside, 7 inches high and 4 inches in diameter. The vessel shall be sealed with a lid lined with nonconducting material having three holes, one at the center for a thermometer, and two diametrically opposite near the rim. These holes shall receive copper tubes of 1/2 inch diameter so arranged that when the lid is in place, one tube enters the oven to a depth of

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6 inches while the other rises to an equal height above the lid. These tubes shall assure a constant draft of air through the instrument. On a common vertical axis with the central hole, a cylinder of wire gauze, 6 inches long and 1½ inches in diameter shall be supported within the oven. This apparatus shall be used for the test specified below to determine compliance with 3.14.

4.14.2 Procedure.-A 12 ± 1 gm portion of the sample shall be cut into ¼ inch to ½ inch squares, and placed within the wire cylinder, compressing it to occupy the upper 4¼ inches. The thermometer shall be arranged so that the bulb is in the center of this mass. The water shall be heated to the boiling point, and the cylinder and thermometer introduced, allowing the latter to protrude through a cork placed in the central hole in the lid. The water shall be allowed to boil and the temperature recorded at frequent intervals for 1½ hours, to determine compliance with 3.14. The material shall be considered to fail with respect to the presence of objectionable sizing if the fabric attains a temperature of 100°C within 1 hour, or if it reaches a temperature of 120°C within 1½ hours.

4.15 Retests.-If a lot is rejected, the manufacturer shall have the option of reworking the material by a method satisfactory to the bureau or agency concerned, and resubmitting the lot for acceptance. Samples for retest shall be taken from each 500 yards of resubmitted material.

5. PREPARATION FOR DELIVERY

5.1 Packing, labeling and marking.-Packing, labeling, and marking shall be in accordance with Specification MIL-P-10025.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number, and date of this specification.

6.2 A number 21 needle and the following threads have been found satisfactory:

a. Commercial Grade H (14 pound) water-proofed silk, fishing line thread with a left hand machine twist.

b. Cotton thread with a left hand machine twist having a minimum breaking strength of 5 pounds when tested in accordance with Paragraph F-4E of Specification 7-7A and a minimum thickness of 0.017 inch. (16-4) Thread, Cranska Thread Co., Worcester, Mass.

6.3 A Lowinson's Thread Counting Micrometer has been found satisfactory for counting the number of ends on cloth.

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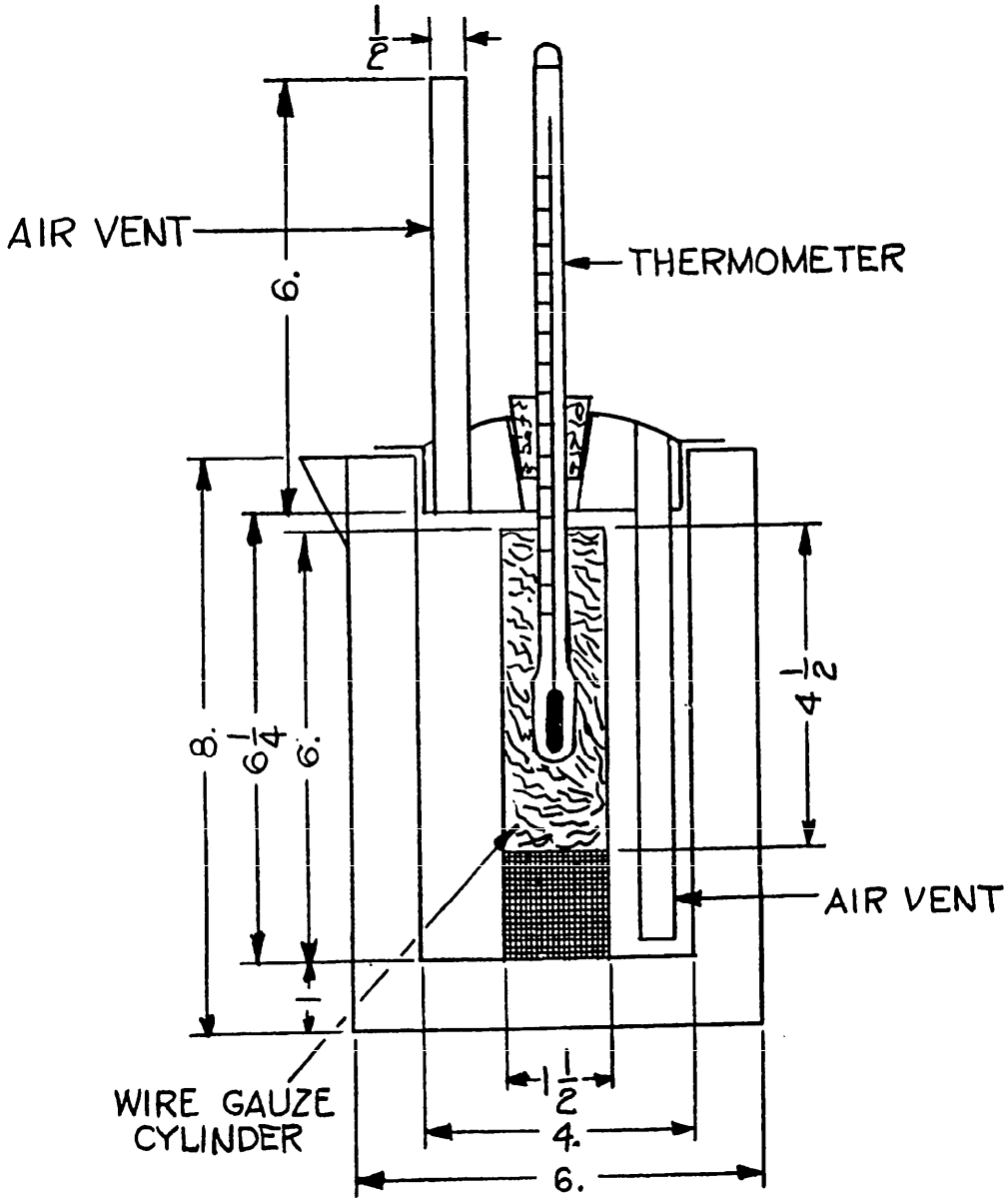
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MACKEY TEST APPARATUS (FOR USE IN SIZING TEST)

FIGURE 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof).*

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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